Relationships and Relationship Sets in DBMS

What is a Relationship?

- A relationship shows an association between entities.
- It explains how entities are connected with each other.
- Relationships are important in the ER model.

Example

- Student enrolls in Course
- Teacher teaches Subject

What is a Relationship Set?

- A collection of similar relationships is called a Relationship Set.
- It contains all relationships of the same type.

Example

If many students enroll in courses:

Student	Course
Ravi	DBMS
Anu	Java

- All these "enrolls" relationships form a Relationship Set.

Important Points About Relationships

Basic Understanding

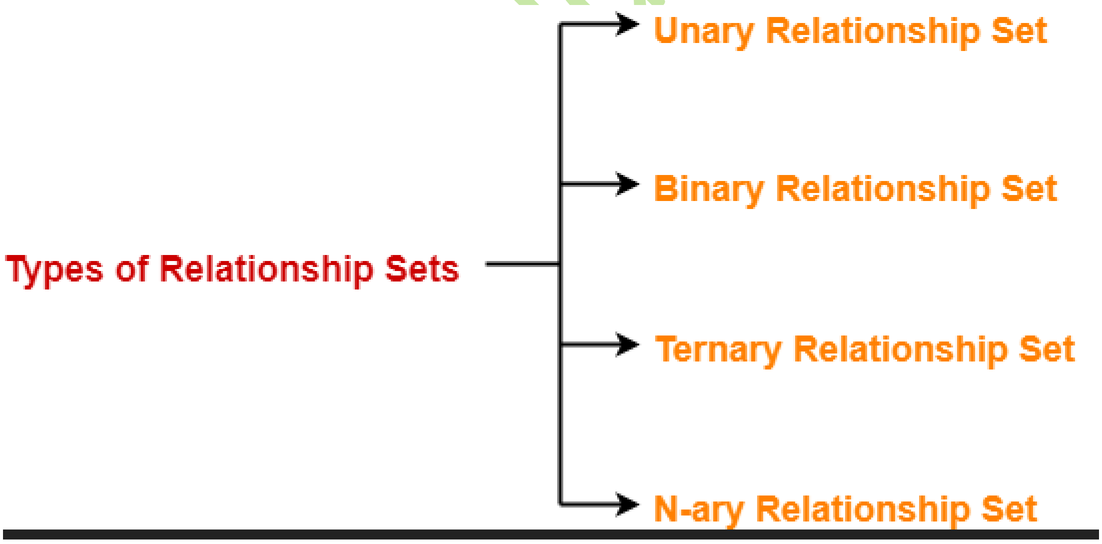
1. Relationships connect entities.
2. Relationships show interaction between entity sets.
3. Relationships are represented using diamonds in ER diagrams.
4. Relationships help organize related data.
5. Relationships improve database structure.
6. A relationship may contain attributes.

7. Relationships help reduce redundancy.
8. Relationships are important in ER modeling.
9. One relationship can connect multiple entities.
10. Relationship sets store all similar relationships.

Examples of Relationships

Entity 1	**Relationship**	**Entity 2**
Student	Enrolls	Course
Teacher	Teaches	Subject
Customer	Purchases	Product
Employee	Works_In	Department

Types of Relationships Based on Degree



1. Unary Relationship

Definition

- Relationship between entities of the same entity set.

Example

- Employee manages Employee

Important Points

11. Also called recursive relationship.
12. Same entity participates multiple times.

2. Binary Relationship

Definition

- Relationship between two entity sets.

Example

- Student enrolls in Course

Important Points

13. Most common relationship type.
14. Connects two different entities.

3. Ternary Relationship

Definition

- Relationship among three entity sets.

Example

- Supplier supplies Product to Store

Important Points

15. Involves three entities together.
16. More complex than binary relationship.

Types of Relationship Mapping

1. One-to-One (1:1)

Definition

- One entity related to only one entity.

Example

- One person has one passport.

Important Points

17. Single entity connected to single entity.
18. Rare in large databases.

2. One-to-Many (1:M)

Definition

- One entity related to many entities.

Example

- One teacher teaches many students.

Important Points

19. One side has single entity.

20. Other side has multiple entities.

3. Many-to-One (M:1)

Definition

- Many entities related to one entity.

Example

- Many students belong to one department.

Important Points

21. Opposite of one-to-many.
22. Common in databases.

4. Many-to-Many (M:N)

Definition

- Many entities related to many entities.

Example

- Students enroll in many courses.

Important Points

23. Most common relationship type.
24. Requires junction/bridge table in relational databases.

Relationship Attributes

25. Relationships can also have attributes.
26. These attributes describe the relationship.

Example

- Enrollment relationship may have:
  - Date
  - Grade

Relationship Set Characteristics

27. Relationship sets connect entity sets.
28. Relationship sets contain relationship instances.
29. Relationship sets help maintain data consistency.
30. Relationship sets are represented by diamonds.
31. Relationships improve data retrieval.
32. Relationships help in query processing.
33. Relationship sets support database integrity.
34. Relationships may be mandatory or optional.
35. Relationship sets can involve strong or weak entities.

ER Diagram Representation

Binary Relationship

[STUDENT] ---- (ENROLLS) ---- [COURSE]

One-to-Many Relationship

[TEACHER] 1 ---- teaches ---- M [STUDENT]

Advantages of Relationships

36. Connects related data
37. Improves database design
38. Reduces data duplication
39. Makes querying easier
40. Supports real-world modeling
41. Helps maintain consistency
42. Improves data integrity
43. Simplifies database management
44. Organizes complex data
45. Helps create meaningful connections

Additional Important Points

46. Relationships are part of ER model.
47. Relationship sets are collections of relationships.
48. Relationship mapping defines connectivity.
49. Relationships are converted into foreign keys in tables.
50. Proper relationships improve database efficiency.

Real-Time Example

College Database

Entity Sets

- Student
- Course
- Faculty

Relationships

- Student enrolls in Course
- Faculty teaches Course

Short Definition for Exams

Relationship:
An association between two or more entities is called a relationship.

Relationship Set:
A collection of similar relationships is called a relationship set.